

Notations :

- 1.Options shown in green color and with ✓ icon are correct.
- 2.Options shown in red color and with ✗ icon are incorrect.

Change Font Color :	No
Change Background Color :	No
Change Theme :	No
Help Button :	No
Show Reports :	No
Show Progress Bar :	No

AI

Section Id :	640653134546
Section Number :	1
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	6
Number of Questions to be attempted :	6
Section Marks :	20
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653374412
Question Shuffling Allowed :	No

Question Number : 1 Question Id : 6406532039983 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : AI: SEARCH METHODS FOR

PROBLEM SOLVING (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE [TOP](#) FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406536642210. ✓ Yes

6406536642211. ✗ No

Sub-Section Number :

2

Sub-Section Id :

640653374413

Question Shuffling Allowed :

No

Question Number : 2 Question Id : 6406532039984 Question Type : MSQ

Correct Marks : 0 Max. Selectable Options : 0

Question Label : Multiple Select Question

**ASK FOR PRINTED
GRAPH SHEETS
8 PAGES TWO-SIDE PRINT**

Options :

6406536642212. ✓ Printed graph sheets were provided on time.

6406536642213. ✗ Printed graph sheets were provided late.

6406536642214. ✗ Printed graph sheets were not provided.

6406536642215. ✓ I used the graph sheets.

6406536642216. ✗ I did not use graph sheets.

Sub-Section Number :

3

Sub-Section Id :

640653374414

Question Shuffling Allowed :

No

Question Id : 6406532039985 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (3 to 9)

Question Label : Comprehension

SEARCH

The figure shows a map on a grid where each tile is 1x1 in size.

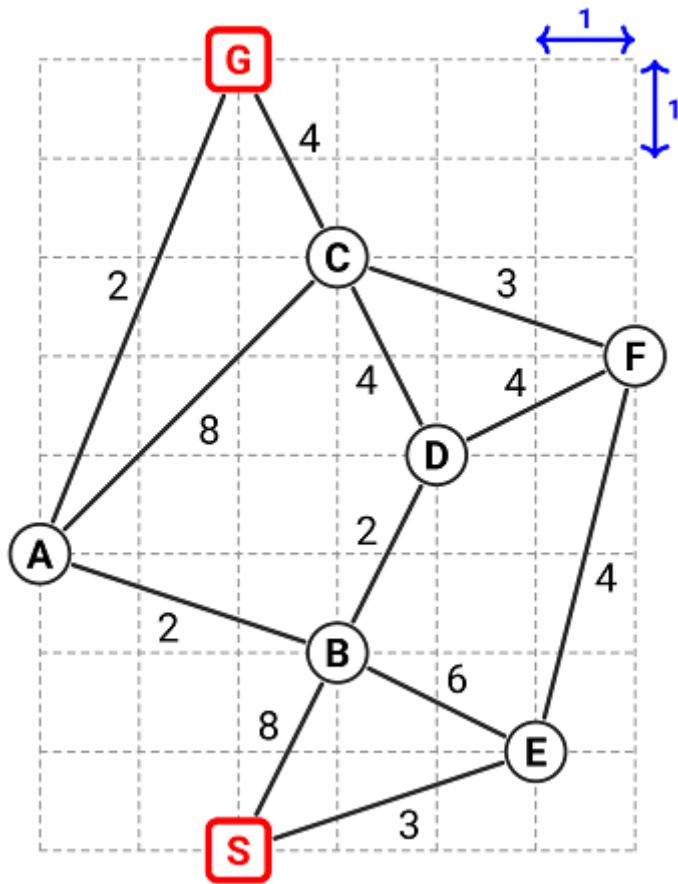
All locations are at grid points.

The start node is S, and the goal node is G.

MoveGen returns neighbours in alphabetical order.

Use Manhattan distance as the heuristic function.

Tie-breaker: Use alphabetical order to break ties.



Emulate A*, WA* (w=2) and Branch-and-Bound on the given map, then answer the sub-questions.

Sub questions

Question Number : 3 Question Id : 6406532039986 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

In the map, S is the first node to be refined, determine the next 3 nodes (from the 2nd to 4th node) refined by A*. Enter the nodes in the order they are refined.

Enter a comma separated list of node labels.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

E,F,C

Question Number : 4 Question Id : 6406532039987 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

What is the final path found by A* ?

Enter the path as a comma separated list.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: S,X,Y,G

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

S,E,F,C,G

Question Number : 5 Question Id : 6406532039988 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

For w=2, what is the final path found by WA* ?

Enter the path as a comma separated list.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: S,X,Y,G

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

S,B,D,C,G

Question Number : 6 Question Id : 6406532039989 Question Type : SA Keyboard Layout :

Inscript

Correct Marks : 1

Question Label : Short Answer Question

What is the cost of the path found by Branch-and-Bound algorithm?

Enter a natural number.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: 42

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

12

Question Number : 7 Question Id : 6406532039990 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

For the given map, which algorithms find a path that is also an optimal path?

Options :

6406536642221. ✖ A*

6406536642222. ✔ Branch and Bound

6406536642223. ✔ Breadth First Search

6406536642224. ✖ Hill Climbing

6406536642225. ✖ WA* (w=2)

Question Number : 8 Question Id : 6406532039991 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Is the heuristic admissible?

Options :

6406536642226. ✖ Yes

6406536642227. ✔ No

Question Number : 9 Question Id : 6406532039992 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

A* algorithm calls MoveGen function, then drops neighbours already present in either OPEN or CLOSED, only then adds remaining neighbours to OPEN.

Options :

6406536642228. ✖ True

6406536642229. ✓ False

Sub-Section Number :

4

Sub-Section Id :

640653374415

Question Shuffling Allowed :

No

Question Id : 6406532039993 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (10 to 12)

Question Label : Comprehension

TSP

Distance matrix for 6 cities (A to F) and sorted segments are provided in the figure. For each city the distances to other cities are listed in ascending order. For example, Column 1 shows the distance from A to D as 14, A to B as 26 and so on.

A	B	C	D	E	F
D : 14	C : 10	B : 10	A : 14	B : 20	A : 32
B : 26	E : 20	E : 24	B : 30	C : 24	D : 36
F : 32	A : 26	A : 44	F : 36	A : 40	B : 38
E : 40	D : 30	D : 56	C : 56	F : 86	C : 64
C : 44	F : 38	F : 64	E : 92	D : 92	E : 86

BC	10	BD	30	AC	44
AD	14	AF	32	CD	56
BE	20	DF	36	CF	64
CE	24	BF	38	EF	86
AB	26	AE	40	DE	92

Note 1: A segment is a two-way edge between a pair of cities.

Note 2: In TSP BnB, after adding or dropping a permanent-segment, first, infer new (included/excluded) permanent-segments, then compute the lower bound.

Use the above information to answer the sub-questions.

Sub questions

Question Number : 10 Question Id : 6406532039994 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

According to the TSP BnB algorithm covered in the lecture, select the first segment XY to refine the root node S0, then compute the lower bound cost of the node (S0, ~XY) that permanently excludes XY. Enter the lower bound cost in the textbox.

Enter a natural number.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: 42

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

155

Question Number : 11 Question Id : 6406532039995 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

How many tours are represented by the node (S0, BC, AD, BE, ~AB, AF)?

Enter a natural number.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: 42

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

2

Question Number : 12 Question Id : 6406532039996 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

Infer all the permanently included and permanently excluded segments in the node (S0, BC, AD, BE, ~AB, AF). Enter the total number of permanently included segments followed by the total number of permanently excluded segments as a comma separated list.

Enter two natural numbers as a comma separated list.
NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.
Answer format: 17,42

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

4,7

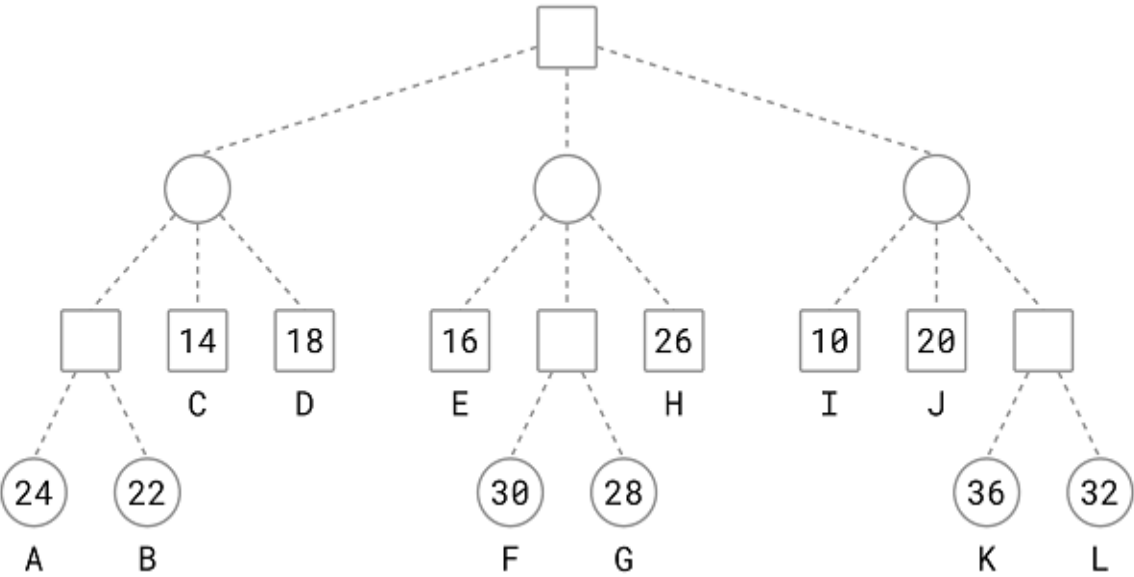
Sub-Section Number :	5
Sub-Section Id :	640653374416
Question Shuffling Allowed :	No

Question Id : 6406532039997 **Question Type :** COMPREHENSION **Sub Question Shuffling Allowed :** No **Group Comprehension Questions :** No **Question Pattern Type :** NonMatrix

Question Numbers : (13 to 17)

Question Label : Comprehension

GAMES



Based on the above data, answer the given subquestions.

Sub questions

Question Number : 13 **Question Id :** 6406532039998 **Question Type :** MSQ

Correct Marks : 1 **Max. Selectable Options :** 0

Question Label : Multiple Select Question

What defines the best strategy for the MAX player?

Options :

6406536642233. ✓ It is a strategy for the MAX player.

6406536642234. ✓ It yields the best value for the MAX player.

6406536642235. ✓ It admits only perfect play.

6406536642236. ✗ It always wins the game for the MAX player.

Question Number : 14 Question Id : 6406532039999 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

For the given game tree, list the leaf nodes in the best strategy.

Enter the node labels as a comma separated list in ASCENDING order.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

E,F,H

Question Number : 15 Question Id : 6406532040000 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

List the leaf nodes pruned by Alpha-Beta algorithm.

Enter the node labels as a comma separated list in ASCENDING order.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

G,J,K,L

Question Number : 16 Question Id : 6406532040001 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

Solve the game tree using SSS* algorithm. List the leaf nodes (that are not in the initial cluster) that are assigned SOLVED status.

Enter the node labels as a comma separated list in ASCENDING order.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: X,Y,Z

Response Type : Alphanumeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Answers Case Sensitive : No

Text Areas : PlainText

Possible Answers :

C,F,H

Question Number : 17 **Question Id** : 6406532040002 **Question Type** : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

Alpha value is refined by Alpha-Beta algorithm whenever _____ .

Options :

6406536642240. ✖ eval(child) <= Alpha

6406536642241. ✔ Alpha < eval(child) < Beta

6406536642242. ✖ Beta <= eval(child)

6406536642243. ✖ Alpha = eval(child) = Beta

Question Id : 6406532040003 **Question Type** : COMPREHENSION Sub Question Shuffling

Allowed : No **Group Comprehension Questions** : No **Question Pattern Type** : NonMatrix

Question Numbers : (18 to 22)

Question Label : Comprehension

AUTOMATED PLANNING

The domain description of Blocks World with a one-armed robot is given below. The same domain description used in assignments.

PREDICATES

armEmpty The arm is empty.
holding(X) The arm is holding X.
onTable(X) X is on the table.
clear(X) X has nothing above it.
on(X,Y) X is directly placed on Y.

OPERATORS

Pickup(X): pick up X from the table.
 Preconditions: { armEmpty, clear(X), onTable(X) }
 Add Effects : { holding(X) }
 Del Effects : { armEmpty, onTable(X) }

Putdown(X): place X on the table.
 Preconditions: { holding(X) }
 Add Effects : { armEmpty, onTable(X) }
 Del Effects : { holding(X) }

Unstack(X,Y): pick up X that is directly on Y.
 Preconditions: { armEmpty, clear(X), on(X,Y) }
 Add Effects : { clear(Y), holding(X) }
 Del Effects : { armempty, on(X,Y) }

Stack(X,Y): place X directly on Y.
 Preconditions: { holding(X), clear(Y) }
 Add Effects : { armEmpty, on(X,Y) }
 Del Effects : { holding(X), clear(Y) }

Tie-breaker 1: When choosing actions non-deterministically, choose actions that lead to a plan. Ignore actions that lead to dead-ends or cycles.

Tie-breaker 2: Treat the goal description, preconditions and effects as lists that are accessed from left to right.

Tie-breaker 3: When list elements are pushed one by one to a stack, the last element in the list will be at the top of the stack.

A planning problem is given below, find a plan using operators and predicates provided in the blocks-world domain.

Start: { armEmpty,
 onTable(B), onTable(C), onTable(E),
 on(A,B), on(D,C),
 clear(A), clear(D), clear(E)
 }

Goal Description: { on(B,C), on(D,E), on(A,B) }

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 18 Question Id : 6406532040004 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following are **applicable** actions for the given planning problem?

Options :

6406536642244. ✓ Pickup(E)

6406536642245. ✗ Stack(A,B)

6406536642246. ✗ Stack(B,C)

6406536642247. ✗ Stack(D,E)

6406536642248. ✓ Unstack(A,B)

6406536642249. ✓ Unstack(D,C)

Question Number : 19 Question Id : 6406532040005 Question Type : MSQ

Correct Marks : 1 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following are **relevant** actions for the given planning problem?

Options :

6406536642250. ✗ Pickup(E)

6406536642251. ✓ Stack(A,B)

6406536642252. ✓ Stack(B,C)

6406536642253. ✓ Stack(D,E)

6406536642254. ✗ Unstack(A,B)

6406536642255. ✗ Unstack(D,C)

Question Number : 20 Question Id : 6406532040006 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

For the given goal description, as per the tie breaking rules, which of the following is the first action popped out of the stack by Goal Stack Planning?

Options :

6406536642256. ✗ Stack(A,B)

6406536642257. ✗ Stack(B,C)

6406536642258. ✗ Stack(D,E)

6406536642259. ✗ Unstack(A,B)

6406536642260. ✓ Unstack(D,C)

Question Number : 21 Question Id : 6406532040007 Question Type : SA Keyboard Layout : Inscript

Correct Marks : 1

Question Label : Short Answer Question

For the given goal description, as per the tie breaking rules, Goal Stack Planning will push the goal description (compound goal) to the stack _____ time(s).

Enter a natural number.

NO SPACES, TABS, DOTS, BRACKETS OR EXTRANEIOUS CHARACTERS.

Answer format: 42

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

2

Question Number : 22 Question Id : 6406532040008 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

While working on a new subgoal, if Goal Stack Planning breaks a previously completed subgoal then it will immediately terminate and report failure.

Options :

6406536642262. ✖ True

6406536642263. ✔ False